

Isolation of heavy metal-resistant *Staphylococcus epidermidis* strain TWSL_22 and evaluation of heavy metal bioremediation potential of recombinant *E. coli* cloned with isolated *cadD*

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Abstract

A heavy metal-resistant bacterial strain, TWSL_22 was isolated from an industrial effluent sample and tested for heavy metal tolerance and resistance. The strain was molecularly characterized as *Staphylococcus epidermidis* based on 16S rDNA gene analysis and the sequence was deposited in the NCBI repository (accession number KT184893.1). Metal removal activity ($P < .001$) of TWSL_22 was $99.99 \pm 0.001\%$, $74.43 \pm 2.51\%$, and $51.16 \pm 4.17\%$ for Cd, Pb, and Cu, respectively. Highest MIC was observed for Cd. Antibiotic susceptibility assays revealed the strain TWSL_22 to be resistant to several antibiotics. The strain was screened for possible heavy metal-resistant genes and presence of *cadA*, *copA*, and *cadD* was confirmed by PCR. A DNA fragment containing complete sequence of *cadD* (618 bp) was isolated and cloned into pET 21a(+), transformed into *E. coli* BL21 and designated as *E. coli/cadDET*. *E. coli/cadDET* showed high metal tolerance capacity and could remove over 82% of heavy metals (Zn^{2+} , Cd^{2+} , Cu^{2+} , and Cr^{3+}) in the industrial effluent.

Keywords: heavy metals, *Staphylococcus*, *cadD*, industrial effluent, bioremediation capacity

Introduction

Industrialization is a major reason for heavy metal contaminations in water as most industrial wastes contain high concentrations of “Potentially Toxic Elements” (PTE) (Taghipour and Jalali 2018). PTE is an alternative to the term “heavy metal.” Toxic and essential metals, and their compounds may also be incorporated into this category (Bhat and Khan 2011, Gholizadeh et al. 2015, Bilal et al. 2018, Pourret and Hursthouse 2019). In recent years, there has been an increasing ecological and global public health concern associated with environmental contamination by these PTE (Masindi and Muedi 2018). To protect natural habitats from these toxic pollutants, country-wise and worldwide regulations have been imposed on discharge of waste. In Sri Lanka, industries are required to obtain an Environmental Protection License from the Central Environment Authority (CEA) prior to commencement of their operations. Therefore, the limits for Cd, Cu, and Pb ions in industrial waste, discharged to inland surface water in Sri Lanka should be less than or equal to 0.1 mg/l, 3.0 mg/l, and 0.1 mg/l, respectively (National environmental Act, number 47 of 1980, 2008). Thus, proper attention need be given to heavy metal remediation technologies.

Remediation can be accomplished through physical, chemical, and biological technologies. Biological technologies are developed using naturally evolved cellular mechanisms to elimi-

nate the toxicity of pollutants in an eco-friendly manner. Plants, fungi, bacteria, or their products are involved in the metal detoxification processes (Dixit et al. 2015). In bioremediation, biological mechanisms inherent in microbes and plants are used (Ojuederie and Babalola 2017). Bacteria living in contaminated sites can withstand high concentrations of heavy metals due to either their capability of biosorption, bioaccumulation, biotransformation, or biomineralization (Dixit et al. 2015). To accomplish these processes, bacteria have evolved several metal tolerance mechanisms (extracellular/intracellular sequestration; Gupta and Diwan 2016), metal exclusion by permeable barriers (Qin et al. 2017), metal efflux mechanisms (Banerjee 2003, Silver 1998), and enzymatic detoxification (Essa et al. 2017).

Several studies have been reported on the heavy metal tolerance of *Pseudomonas* spp., *Bacillus* spp., *Staphylococcus* spp., *Enterococcus* spp., and *Synechococcus* sp., and their metal-resistant genes (*merA*, *merB*, *copA*, *copB*, *copZ*, *cadA*, *cadC*, *cadD*, *smtAB*, *arsA*, *arsB*, *silS*, and *silR*) (Chaouni et al. 1996, Silver 1998, Crupper et al. 1999, Yu et al. 2020, Gutierrez et al. 2019, Kumari et al. 2021b). The membrane proteins or P-Type ATPases family transporters are encoded by genes such as *copA*, *cadA*, *pbrA*, *arsA*, and *arsB* (Nucifora et al. 1989, Hu et al. 2005, Arguello et al. 2013), and are mainly involved in discharging metals (Arguello et al. 2013). *CadD* protein shows a higher degree of homology to *CadB*, which is also a metal

binding transmembrane protein that does not promote cation efflux (Crupper et al. 1999, Hoogewerf et al. 2015).

Current research on bioremediation includes the development of metal specific biotechnological tools such as biosensors and biosorbents using bacterial metal resistance genes (Gui et al. 2017). This study was carried out to in vitro evaluate the metal tolerance and bioremediation capacity of recombinant *Escherichia coli* strain BL21 containing *cadD* of *Staphylococcus epidermidis* TWSL_22. The gene *cadD* was first identified and studied from *Staphylococcus aureus* plasmid pRW001, which showed a high degree of sequence similarity to the *cadB*-like gene that encodes Cd resistance CadB in *S. lugdunensis*, which is also a metal binding transmembrane protein that does not promote cation efflux (Perry and Silver 1982, Crupper et al. 1999, Hoogewerf et al. 2015).

Materials and methods

Sample collection and isolation of heavy metal-resistant bacteria

An effluent sample was collected from a textile dyeing industry in the Western province, Sri Lanka. Initially, the temperature, pH, color, and odor of the sample were recorded. Presence and concentrations of the metals in the effluent were determined using an atomic absorption spectrophotometer (AAS) (GCB 932 Plus). To screen for possible metal tolerant bacterial strains, present in the effluent, 50 µl from prepared dilutions (10^{-1} – 10^{-6}) were separately spread on LB agar plates. After incubation (overnight, 37°C), well-isolated distinct bacterial colonies were restreaked and a few isolated bacterial strains were screened for heavy metal resistivity by observing their growth on multimetal enriched LB agar plates [Cd^{2+} , Pb^{2+} , and Cu^{2+} (20 mg/l)] for 2 days (Benson 2002, Kumari et al. 2021a,b). A bacterial strain designated TWSL_22, which grew well was selected as a most promising strain for further analysis.

Determination of heavy metal removal capacity and heavy metal tolerance ability of TWSL_22, using metal-spiked cultures

A total of two aliquots of LB broth (100 ml) were separately spiked with heavy metals [Cd^{2+} , Pb^{2+} , and Cu^{2+} (1 mg/l)] and one was inoculated with 50 µl ($\sim 2 \times 10^7$ CFU/ml) of an overnight grown TWSL_22 culture. Other one (without the bacterial inoculum) was used as the abiotic control. The growth of TWSL_22, with time was determined by measuring the optical density at 600 nm (OD_{600}) using a scanning UV-VIS spectrophotometer [Spectro UV-VIS Dual Beam (Split), Helios Alpha Thermo Fisher Scientific] at time intervals of 1, 3, 5, and 7 days. Minimum inhibitory concentration (MIC) was also determined by spreading 10 µl of an overnight grown culture on LB agar plates supplemented with metal ions (Cd^{2+} , Pb^{2+} , and Cu^{2+}) at different concentrations (1–1000 mg/l). The concentration at which no growth was observed on the third day was considered as the MIC (Kumari et al. 2021b, Kowalska-Krochmal et al. 2021). In order to determine heavy metal removal capacity of the strain, concentrations of metals in the cultures were also determined at the same predefined time intervals using AAS (GCB 932 Plus). Aliquots from above samples were filter sterilized (0.25 µm) prior to analysis. The differences between the initial and final concentrations of heavy metals in the media (Radhika et al. 2006) were determined and bioremoval percentage (BR%) of metal ions was calculated using equation (1) (Kumari et al. 2021b).

$$\text{BR}\% = [(C_0 - C_t)/C_0] \times 100, \quad (1)$$

where C_0 and C_t are metal ion concentrations at a time, 0 min and at time interval t , respectively.

Characterization of the bacterial strain

The colony and cellular morphology, Gram staining, and motility of pure colonies of TWSL_22 were determined using a compound light microscope (Olympus CKX41) (Benson 2002). Biochemical tests (methyl red test, starch hydrolysis test, urease test, catalase test, Simmons citrate agar test, coagulase test, gelatin hydrolysis test, and MacConkey agar test) were also carried out (Benson 2002).

Molecular characterization of TWSL_22 was carried out by sequencing of the 16S rDNA gene. Briefly, genomic DNA was extracted using the optimized guanidine thiocyanate-based method (Pitcher et al. 1989, Kumari et al. 2021b) and the 16S rDNA gene was amplified using the primer pair fd1/rD1 (Weisburg et al. 1990). The amplified product (~ 1500 bp) was sequenced (Macrogen, Republic of Korea) and analyzed. The sequence was compared with DNA sequences available in the National Center for Biotechnology Information (NCBI) database using Basic Local Alignment Search Tool (BLAST). Multiple sequence alignment was carried out using Clustal W software and a phylogenetic tree was constructed using the neighbor-joining method based on the Kimura 2-parameter model, using the MEGA X software (Kimura 1980, Saitou and Nei 1987, Kumar et al. 2018).

The antibiotic susceptibility of the TWSL_22 was also determined using the Kirby Bauer disk diffusion susceptibility test method (Bauer et al. 1959, CLSL 30th edition, 2020). Ampicillin, amoxicillin, ciprofloxacin, vancomycin, cefotaxime, erythromycin, and gentamycin were selected for the assay, considering the antibiotic resistivity of heavy metal-resistant *Staphylococcus* spp. (Groves et al. 1975, Berg et al. 2005). Briefly, overnight grown, well isolated colonies were resuspended in sterile saline solution (0.9% w/v NaCl) until the turbidity reach up to 0.5 McFarland standard ($\sim 1.5 \times 10^8$ CFU/ml). This was used in disk diffusion assays and the diameters of inhibition zones (mean value of three replicates) were measured using a ruler.

Identification of heavy metal-resistant genes in TWSL_22

Presence of the possible genes (*cadD*, *cadA*, and *copA*) were identified through a literature survey and gene specific primers, Cad5, Cad3, and Cop2 for PCR amplification of *cadD*, *cadA*, and *copA* genes were designed, respectively (Table 1). The partial sequence of *cadD* was PCR amplified using GoTaq® DNA Polymerase [initial denaturation (95°C, 2 min), followed by 35 cycles of denaturation (94°C, 30 s), annealing (52°C, 1 min), extension (72°C, 1 min), followed by a final extension (72°C, 3 min)]. *cadD*, which encodes the heavy metal-resistant CadD was selected for the further analysis. The PCR product of *cadD* was gel purified and sequenced (Macrogen). The partial nucleotide sequence was analyzed using BioEdit software and BLAST. Sequence homology was further confirmed by constructing a phylogenetic tree using the Maximum-Likelihood method and the Jukes-Cantor model (Jukes and Cantor 1969).

Isolation and amplification of the complete coding sequence of *cadD*

Complete *cadD* was PCR amplified from genomic DNA of TWSL_22 [initial denaturation at 95°C (2 min) followed by 35 cycles of denaturation at 94°C (40 s), annealing at 52°C (1 min), extension at 72°C (2 min), followed by a final extension at 72°C (15 min)] using the designed primers, SD-EF and SE-XHR (Table 1). The PCR

Table 1. Primers used in the study

Target DNA sequence	Primer details			Expected size of the PCR product (bp)	Reference
	Primer name	Oligonucleotide sequence (5'→3')	*T _{an} °C		
16s rDNA of TWSL_22	fD1 (Forward)	AGAGTTTGATCCTGGCTCAG	57.8	~1500	Weisburg et al. (1990)
Partial gene sequence of <i>cadD</i>	rD1 (Reverse)	AAGGAGGTGATCCAGCC	57	536	In this study
	Cad5 (Forward)	CAGACTATTGTAACCTGCTGCT	52		
Partial gene sequence of <i>cadA</i>	Cad5 (Reverse)	AACAGCCATAATCCAACGAC	56	580	In this study
	Cad3 (Forward)	TCTACTCCAATCTCGATTGTCTC			
Partial gene sequence of <i>copA</i>	Cad3 (Reverse)	GTACCTTGATTATCACCTGTCAG	49	724	In this study
	Cop2 (Forward)	GTAATGGGAATGACGTGTGCAG			
Complete gene sequence of <i>cadD</i>	Cop2 (Reverse)	TCACCTGTCAACATTGACTCGTC	50	~633	In this study
	SD-EF (Forward)	GAATTCATGATTCAGACTATTGTAACCTG			
	SE-XHR (Reverse)	CTCGAGAAATATAAATCCTAAATTTGTTTG			

*T_{an} – annealing temperature was optimized for each primer pair by performing gradient PCR.

product was then gel purified and sequenced (Macrogen). Nucleotide sequence was analyzed using BioEdit software v7.1.11. and BLAST. The sequence was then deposited in GenBank (accession number: MK900461.1).

Construction of a recombinant bacterial strain with *cadD* (*E. coli/cadDET*)

The confirmed *cadD* DNA sequence was cloned into pGEM®-T Easy vector (TA cloning) and transformed into *E. coli* JM109 competent cells (CaCl₂ method). Positive recombinant clones were screened (colony PCR) and confirmed by sequencing (Macrogen). Recombinant plasmid DNA was then isolated and digested with EcoRI and XhoI restriction enzymes and ligated with modified pET-21a (+) expression vector and transformed into *E. coli* BL21 (DE3) competent cells (CaCl₂ method). Positive recombinants were selected on LB agar plates containing ampicillin (100 µg/ml)/IPTG (1 mM) and then confirmed by colony PCR.

Analysis of industrial effluent for the bioremediation assay with *E. coli/cadDET*

The pH of the collected industrial effluent from a paint industry in Colombo, Sri Lanka was measured and the initial color and odor were noted. It was sterilized prior to analysis by centrifuging (3500 rpm, 30 min) and filtering through a membrane filter (0.25 µm). Initial heavy metal ion concentrations were determined using microwave plasma atomic emission spectrometry (MP-AES). As the initial heavy metal concentrations were found to be at low ppb levels, it was spiked with 10 ppm of each of the heavy metal ions found to be present in the effluent. The filtered effluent sample was then checked for the presence of bacteria by spread plating. Absence of colonies after 48 h growth was considered as a confirmation for the absence of any bacteria in the filtered heavy metal ion-spiked effluent.

Determination of heavy metal tolerance ability and bioremediation potential of recombinant *E. coli/cadDET*

Sterilized effluent (150 ml) was inoculated with overnight grown recombinant *E. coli* BL 21 strain (50 µl; ~2 × 10⁸ CFU/ml) harboring *cadD*. Uninoculated effluent was used as the abiotic control. After a 3-h incubation (37°C) while shaking, the inoculated sample was induced with 150 µl of 1 M IPTG and incubation was continued. The growth of the bacterial strain with time was determined by measuring the optical density at 600 nm (OD₆₀₀) using a scanning UV-VIS spectrometer continuously for 5 days. The cell counts of

the cultures were then calculated (equation 2) (Beal et al. 2003). To determine the exact colony forming units, 100 µl from the culture was plated on LB agar plates using spread plating technique each day (5 days) and incubated (overnight, 37°C).

$$\text{Cell count} = \text{OD}_{600} \times 8 \times 10^8 \text{ cell/ml.} \quad (2)$$

To determine the heavy metal bioremediation potential of *E. coli/cadDET*, LB broth containing ampicillin (100 µg/ml) and bacterial-free heavy metals-spiked industrial effluent (150 ml) was inoculated with *E. coli/cadDET* (50 µl; ~2 × 10⁷ CFU/ml), and effluent with no inoculum was used as the abiotic control. Sample with the recombinant bacteria was induced with 150 µl of 1 M IPTG and the incubation was continued. Aliquots (1 ml) were withdrawn every day (days 01–05) and diluted aliquots [(10⁻¹ v/v) using sterilized deionized water] were used to determine the concentrations of Cu²⁺, Cd²⁺, Zn²⁺, Pb²⁺, and Cr³⁺ metal ions using MP-AES. The reduced concentrations of heavy metals were considered as the difference between the initial and final concentration in the culture. It was assumed that the reduction in metal ion concentration was only due to the bioremediation activity of the *E. coli/cadDET*. BR% of the strain was calculated (equation 3) (Kumari et al. 2021b) as follows:

$$\text{Percentage of heavy metal bioremediation} = \frac{\text{Heavy metal removed (ppm)}}{\text{Initial heavy metal concentration (ppm)}} \times 100. \quad (3)$$

Isolation of CadD protein

A bacterial colony of *E. coli/cadDET* was inoculated into LB broth (5 ml) containing ampicillin (100 µg/ml) and incubated overnight (37°C, 150 rpm). Overnight culture (2 ml) was transferred to 250 ml of fresh LB broth containing ampicillin (100 µg/ml) and incubated until the bacterial cell growth reached an OD₆₀₀ of 0.3–0.5. An aliquot (5 ml) was separated and the remaining culture was induced with 290 µl of 0.5 M IPTG. Following a 2-h incubation, both induced and uninduced cells were harvested (3500 rpm, 15 min, 4°C). Cell pellets were suspended in cell lysis buffer (pH 7.4) (8.0 ml for the induced sample and 192 µl for the uninduced sample) followed by addition of lysozyme (0.04 mg/ml). Following the incubation on ice (30 min), both cell suspensions were sonicated on ice using 10 s bursts with a 10-s cooling period between each burst for about 2 min. The lysates were then centrifuged (3000 g, 15 min) (Sambrook et al. 2012). Supernatants from both samples were collected separately and analyzed by sodium dodecyl sulfate-polyacrylamide gel electrophoresis (SDS-PAGE).

Statistical analysis

All data sets [means of replicates ($n = 3$), standard deviations (SD)] were statistically analyzed using Graph Pad Prism 8.0.1 software. Group analysis ($\alpha = 0.05$) was carried out using "Two-way ANOVA-multiple comparison" and $P < .05$ was considered statistically significant. The graphs were plotted and error bars indicate $\pm$ SD.

Results and discussion

Sampling, isolation, and screening of heavy metal-resistant bacterial strains

The collected untreated industrial waste water sample was found to contain Cu^{2+} (0.050 mg/l) and, Pb^{2+} and Cd^{2+} ions were not detected as the concentrations were below the limit of detection of MP-AES. The temperature of the sample was 38°C and pH was 8.9. It was dark blue-black in color with a pungent and foul odor. From the pool of bacterial strains grown on the LB agar plates, 10 isolates were recovered as pure colonies. Among them, one bacterial strain, which grew well in the multimetal enriched medium was selected and was designated as TWSL_22.

Heavy metal removal capacity and tolerance ability of TWSL_22

The strain TWSL_22 was found to be resistant to Cd^{2+} , Pb^{2+} , and Cu^{2+} and grew well ($P < .0001$) in metal-spiked media. The growth of the bacterial strain in the presence of metal ions was significantly different ($P < .05$) and showed a steady growth till day 5 (Fig. 1A). The highest growth rate was observed in the presence of Cu^{2+} at the beginning, while the lowest was with Cd^{2+} . On the seventh day, a high growth rate was observed in the presence of Cd^{2+} than the Cu^{2+} and Pb^{2+} (Fig. 1A). The initial stress due to heavy metals resulted in a drop in the growth rate, and the activation of detoxification mechanisms subsequently supported the growth later. The steady growth of TWSL_22 throughout the growth period shows its high tolerance for metal ions. TWSL_22 showed a significantly high resistance for Pb (MIC: 850 mg/l) and MIC values for Cd^{2+} and Cu^{2+} were 25 mg/l and 50 mg/l, respectively. It also showed significant metal removal ability for all the metal ions tested over a 7-day period compared to the control ($P < .05$) ($P < .0001$) (Fig. 1B and C). As the control sample (C), did not show any significant reduction ($P < .01$) of metal ion concentrations, it can be concluded that the decrease in metal ion concentration is solely due to the metal removal ability of the inoculated TWSL_22. The strain TWSL_22 had the ability to remove $\sim 100\%$ of Cd and removed over 50% of all the tested metal ions from their initial concentrations (1 mg/l) on the seventh day after inoculation ($P < .05$) (Fig. 1C). Maximum BR% were achieved on the seventh day, which were $99.99 \pm 0.00\%$, $74.43 \pm 2.51\%$, and $51.16 \pm 4.17\%$ for Cd^{2+} , Pb^{2+} , and Cu^{2+} , respectively. Therefore, it can be concluded that the Cd^{2+} removal mechanism of TWSL_22 gets activated upon exposure to metals. A similar behavior has been reported with a *Staphylococcus* sp. isolated from a heavy metal mining area and it was shown that its ability to cope with metal toxicity was due to intrinsic properties of the bacterium (Escamilla-Rodriguez et al. 2021). Evolved heavy metal resistance had further been observed in *S. aureus* strains, by analyzing metal resistance and viability using metallothionein as metal binding protein. Changes in both biochemical and protein profiles of heavy metal-resistant *S. aureus* was found to be due to modifications in protein expression and inhibition/activation of enzymes by metals (Chudobova et al. 2015).

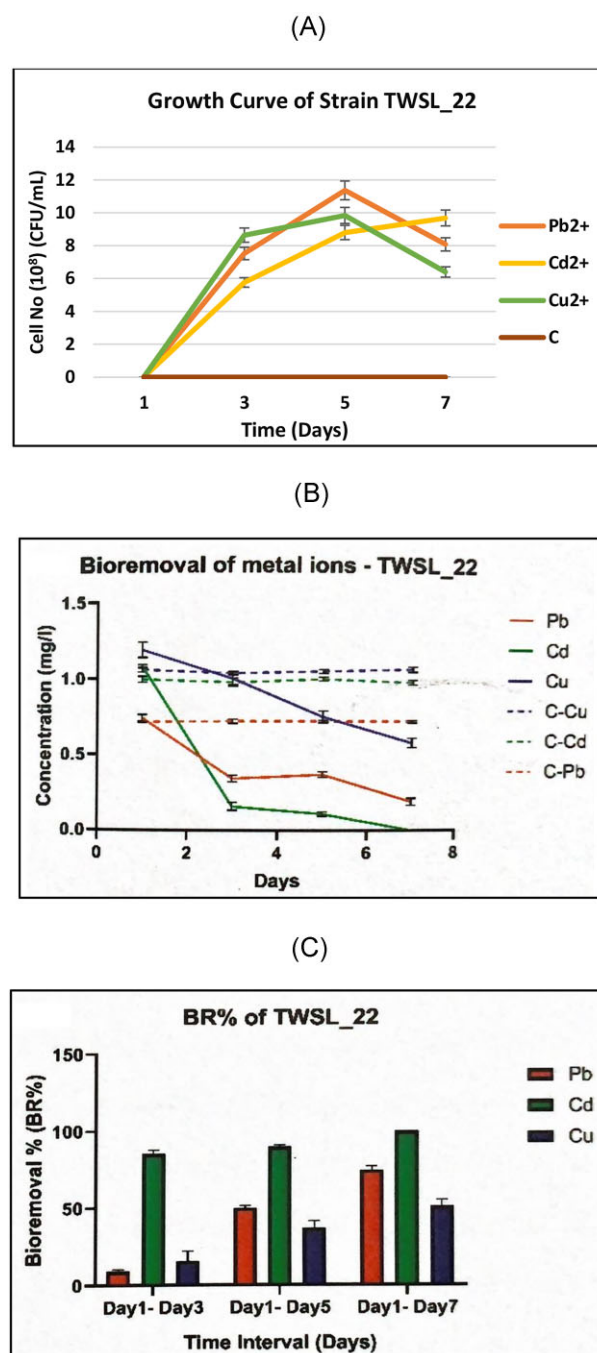


Figure 1. Metal removal ability and tolerance of bacterial strain TWSL_22 during the 7-day period. (A) Growth curve of TWSL_22 bacterial strain from days 1 to 7 in different metal ions (Cd^{2+} , Pb^{2+} , and Cu^{2+})-spiked cultures (mg/l). "C" represents the abiotic control where no inoculum was present and contain LB broth with metals. (B) Heavy metal removal of TWSL_22 with time. (C) Percentage removal of metals with time. Cu^{2+} -, Cd^{2+} -, and Pb^{2+} -enriched media (1 mg/l) were used for the assay and abiotic controls for the respective metal ions (Cu^{2+} , Cd^{2+} , and Pb^{2+}) are labeled as C-Cu, C-Cd, and C-Pb. Error bars represent standard deviations ($n = 3$). Significance of the mean difference was set at $P \geq .05$.

Characterization of the bacterial strain TWSL_22

Microscopic examination of TWSL_22 revealed it to be non-motile, Gram-positive, and irregular (grape-like) clusters of cocci. Colonies were white in color, circular margined, and raised with a shiny/wet surface. The starch hydrolysis test, urease test, and catalase test were positive, which are characteristics of

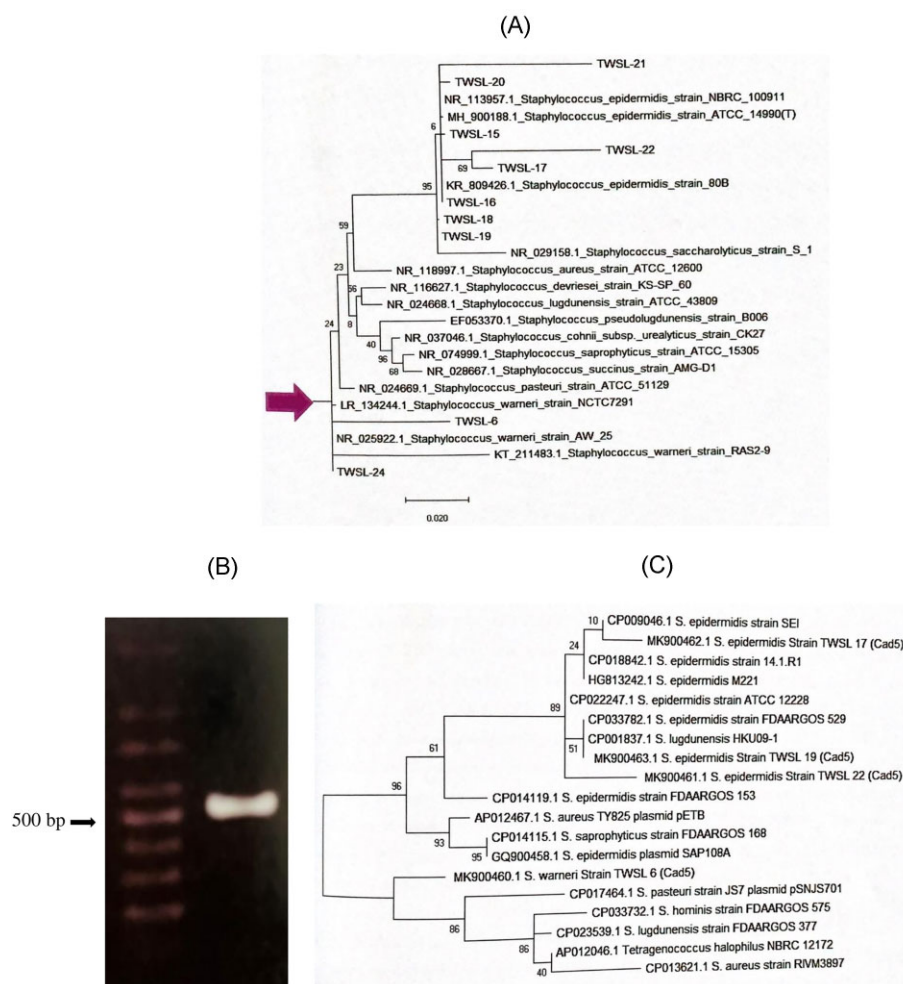


Figure 2. Identification and confirmation of strain TWSL_22 and metal-resistant *cadD* in *S. epidermidis* strain TWSL_22. **(A)** Molecular phylogenetic tree of strain TWSL_22 and other *Staphylococcus* strains based on 16S rDNA sequences [constructed using the neighbor-joining method (Saitou and Nei 1987) based on the Kimura 2-parameter model (Kimura 1980) and the variation among sites was modeled with a gamma distribution (shape parameter = 1). The percentage of replicate trees in which the associated taxa clustered together in the bootstrap test (1000 replicates) are shown next to the branches (Felsenstein 1985). The tree is drawn to scale, with branch lengths measured in the number of substitutions per site. Accession numbers of the sequences used in this study are shown in parenthesis next to the taxon name. Evolutionary analyses were performed using MEGA X software (Kumar et al. 2018)]. **(B)** Amplification of *cadD* partial gene sequence using primer pair Cad5. [Lanes 1–100 bp DNA ladder, Lane 2 – amplified *cadD* (420 bp)]. **(C)** Molecular phylogenetic tree of *cadD* partial sequence [constructed using Maximum-Likelihood method and Hasegawa–Kishino–Yano model for *cadD* sequences. The tree is drawn to scale and branch lengths are in the same units as those of the evolutionary distances used to infer the phylogenetic tree. The evolutionary distances were computed using the Poisson correction method (Nei and Kumar 2000). Evolutionary analyses were performed using MEGA X software (Kumar et al. 2018)].

Staphylococcus sp. and, therefore, easily grouped as *Staphylococci* (Foster 1996, Gnanamani et al. 2017). BLAST analysis of the PCR amplified 16S rDNA sequence (~1500 bp) showed it to be identical (99%) to the 16S rDNA sequence of *S. epidermidis* strain NBRC 100911 (accession number: NR_113957.1) in NCBI GenBank. Phylogenetic analysis (Fig. 2A) identified the strain TWSL_22 as *S. epidermidis*. Hence, the strain TWSL_22 was named as “*Staphylococcus epidermidis* strain TWSL_22” and the 16S rDNA sequence was deposited in the NCBI repository (accession number KT184893.1).

According to the Kirby Baur guidelines, TWSL_22 was found to be resistant to amoxicillin, ampicillin, and vancomycin. Mean diameters of the inhibition zones ($n = 3$) for amoxicillin, ampicillin, cephalexin, ciprofloxacin, erythromycin, gentamycin, and vancomycin were 1.60 ± 0.05 , 1.93 ± 0.08 , 3.07 ± 0.08 , 2.18 ± 0.03 , 2.50 ± 0.05 , 1.68 ± 0.03 , and 1.02 ± 0.03 [(cm), (mean $\pm$ SD)], respectively (Bauer et al. 1959; CLSL 30th edition 2020). It has also been reported that metal-resistant bacteria could develop resis-

tance to many antibiotics in addition to their cross-tolerance to other metal ions such as cadmium, zinc, and mercury (Naik and Dubey 2011) and some *S. aureus* strains were resistant to gentamycin (Ragbetli et al. 2016) and ampicillin (Yilmaz and Aslantas 2017).

Identification of heavy metal resistance genes in TWSL_22 and isolation of the *cadD* gene

Previous reports have indicated the presence of several heavy metal-resistant genes such as *cadA* (~2500 bp), *cadD* (~600 bp), and *copA* (~2300 bp) in *Staphylococcus* spp. (Crupper et al. 1999, Schirawski et al. 2002, Purves et al. 2018). Based on these sequences, gene specific primers were designed (Table 1) and PCR analysis resulted in the amplification of parts of the *cadA* (580 bp), *cadD* (536 bp) (Fig. 2B), and *copA* (811 bp) genes. BLAST analysis revealed *cadD* to be annotated as “cadmium transporter” (*Staphylococcus saprophyticus* strain FDAARGOS; accession

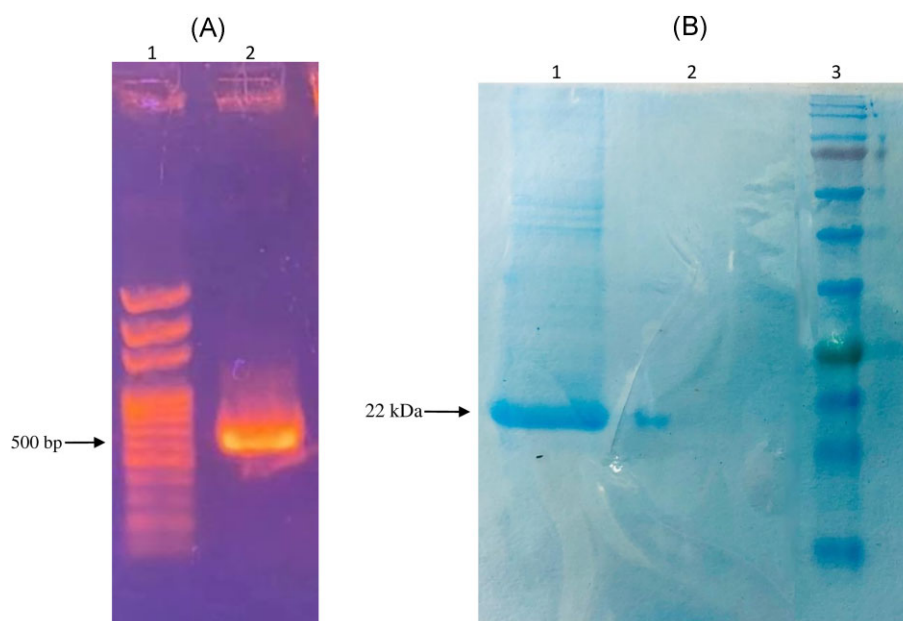


Figure 3. Amplified *cadD* gene and expressed CadD protein. (A) Gel electrophoresis of amplified complete *cadD* [Lane 1: 100 bp ladder; Lane 2: amplified complete gene of *cadD* (619 bp)]. (B): SDS-PAGE showing the expression of CadD in *E. coli/cadDET*. (20% resolving gel) [Lane 1: expressed CadD protein (22 kDa) in IPTG-induced sample; Lane 2: uninduced protein sample; Lane 3: protein marker].

number: CP014115.1), “cadmium-resistant protein” (*S. lugdunensis* HKU09-01; accession number: CP0001837.1, *S. epidermidis* plasmid SAP108A; accession number: GQ900458.1), and “cadmium binding protein” (*S. aureus* TY825 plasmid; accession number: AP012467.1). Evolutionary relationship inferred by phylogenetic analysis of the *cadD* sequence of TWSL_22 is shown in Fig. 2(C). *cadD* partial sequences belonging to *S. epidermidis* species clearly clustered with *cadD* sequences of the same species. One *cadD* partial sequence of *S. epidermidis* TWSL_22 was then deposited in NCBI database (GenBank accession number: MK900461.1).

Complete gene of *cadD* (618 bp/206 aa) was also amplified (Fig. 3A) from the genomic DNA of *S. epidermidis* TWSL_22, using designed SD-EF and SE-XHR primers. The nucleotide sequence obtained was then deposited in NCBI database (accession number: MK900473.1). Cd resistance CadD protein is reported to be either chromosomally or plasmid encoded and mostly found in the family *Staphylococcaceae* (Perry and Silver 1982, Chaoui et al. 1996, Crupper et al. 1999, Chen et al. 2007).

Construction of recombinant *E. coli/cadDET* and expression of CadD

PCR amplified complete *cadD* was cloned into the vector pGMET and then cloned in pET 21a (+) modified vector. The resulting recombinant bacterial strain containing *cadD* was designated as *E. coli/cadDET*. The expression of CadD (22 kDa) by *E. coli/cadDET* was confirmed by SDS-PAGE (Fig. 3B).

Heavy metal removal and tolerance ability of recombinant *E. coli/cadDET* in a metal-spiked industrial effluent

The effluent sample (paint manufacturing company) was dark gray colored, slightly turbid, and odorless. Initial pH was 6.80 and the recommended pH range for growth and recombinant protein expression in *E. coli* BL21 is 6.5 to 7.0 (Wang et al. 2014). Therefore, the initial pH of the effluent was suitable for the growth of the recombinant *E. coli* strains to express metal tolerance pro-

teins. Concentrations of the analyzed metal ions in industrial effluent including Cu^{2+} , Cd^{2+} , Zn^{2+} , Pb^{2+} , and Cr^{3+} were 0.07 ± 0.00 , 0.01 ± 0.00 , 0.05 ± 0.00 , 0.00 ± 0.00 , and 0.00 ± 0.00 (ppm) [mean $\pm$ SEM (standard error of the mean), $n = 3$], respectively. As these concentrations are very low to be used in a bioremediation assay, it had to be spiked with above metal ions to a final concentration 10 ppm each. These low concentrations of heavy metal ions in the effluent indicates that the company from where the effluent sample was obtained comply with the rules and regulations imposed by CEA, Sri Lanka.

Color of the effluent inoculated with *E. coli/cadDET* turned black on day 5 and was significantly different to the color of the control (Fig. 4A) and it was evident that CadD is involved in metal precipitation or accumulation within the cell. It is known that Pb and Cu metal ions form insoluble compounds with sulfides, oxides, or phosphates in this pH range. It has been demonstrated that these insoluble dark particles are due to immobilization of metal ions by extracellular sequestration. Further, analysis has shown that, bacterial spores can entrap the insoluble materials formed (Kumari et al. 2021b). *E. coli/cadDET* was found to be resistant to all the metal ions tested (Cu^{2+} , Cd^{2+} , Zn^{2+} , Pb^{2+} , and Cr^{3+}) and showed an exponential growth till day 5 (Fig. 4B) indicating its metal tolerance.

cadDX operon carrying *cadD* and regulatory *cadX* has been studied previously and *S. aureus* RN4220 transformant carrying *cadDX* in pUR3912 exhibits a high MIC for CdSO_4 (128 mg/l) confirming the involvement of the *cadDX* operon in Cd ion tolerance (Gomez-Sanz et al. 2013). Heavy metal removal of Zn^{2+} , Cd^{2+} , Cu^{2+} , Cr^{3+} , and Pb^{2+} by *E. coli/cadDET* was found to be 95%, 88%, 84%, 86%, and 43% in metal ion-spiked industrial effluents, respectively (Fig. 4C). The reductions of the metal ion concentrations in the control and the nonrecombinant were not significant ($P > .1$). This implies that the reductions of heavy metal ion concentrations of the metal ion-spiked effluent were solely due to the activity of *pcadDET* in *E. coli*. On the third day, concentrations of Zn^{2+} , Cd^{2+} , Cu^{2+} , and Cr^{3+} got reduced by more than 50% of the initial metal concentration. A previous study had demonstrated that the levels of

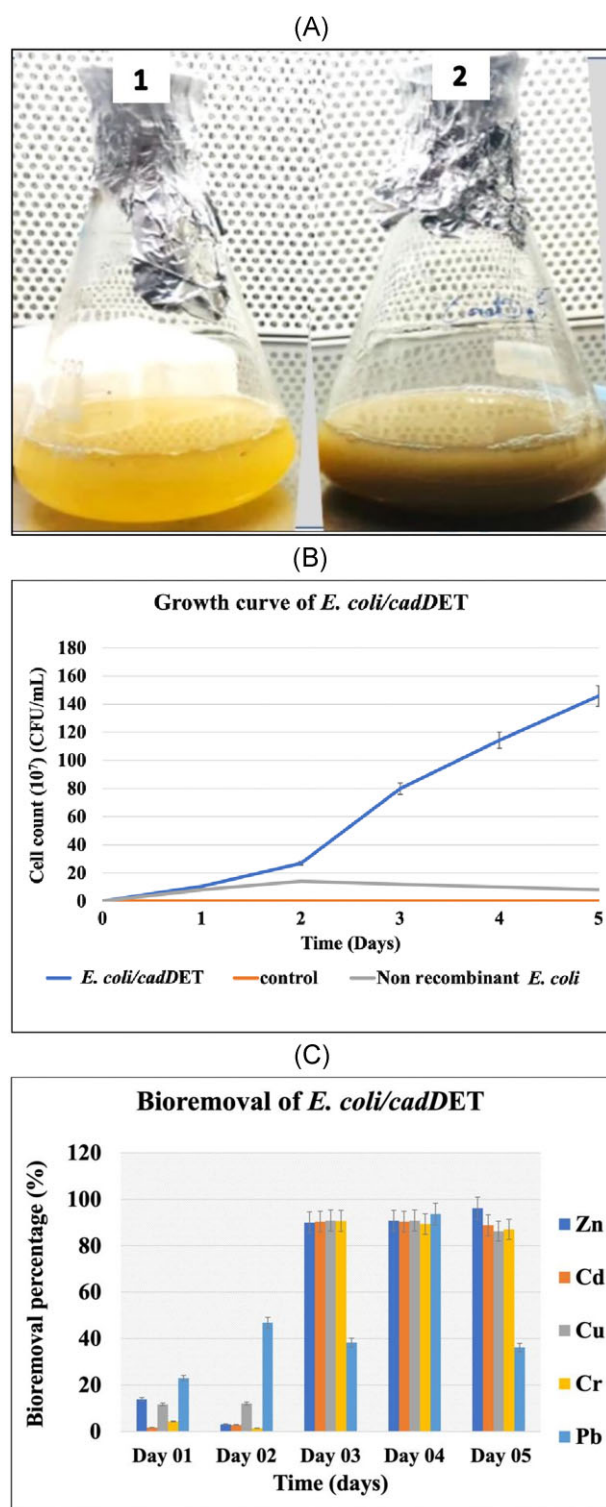


Figure 4. Heavy metal tolerance ability and bioremediation potential of recombinant *E. coli/cadDET*. **(A)** Appearance of the industrial effluent on day 5 [1 – abiotic control (containing metal-spiked industrial effluent without inoculation of recombinant bacteria); 2 – metal-spiked effluent, inoculated with *E. coli/cadDET*]. **(B)** Growth curve of the recombinant *E. coli/cadDET* strain in the metal-spiked effluent. **(C)** Bioremoval of heavy metals during the 5-day period; significance of mean differences were set at $P \geq .05$. LB broth without inoculum was used as the abiotic control. Error bars represent the mean of standard deviation of three replicates ($n = 3$).

resistance for Cd^{2+} and Zn^{2+} by recombinant *E. coli* is more than 9-fold and 6-fold higher, respectively than the *Streptococcus salivarius* wild strain, from which the *cadD* had been isolated (Chen et al. 2007). *E. coli/cadDET* showed considerable bioremoval ability against tested heavy metal ions compared to the abiotic control and the nonrecombinant sample, indicating CadD of TWSL_22 to be involved in the heavy metal removal mechanism.

However, *E. coli/cadDET* did not show a higher removal activity for Pb^{2+} . When considering the cationic radii of metals in this study, Pb^{2+} ions have the highest ionic radius (1.19 Å), which is notably higher than the ionic radius of Cd^{2+} (0.24 Å), the second largest ion among the heavy metals used in the study. The binding sites of metal-binding proteins should be physically compatible with ionic radii of metal ions (Bloom 2019). Further, the sorption maxima is inversely proportional to the hydrated ionic radii of metal ions (Chen et al. 2010). Therefore, it is likely that the Pb^{2+} ions do not fit to the metal binding pockets of the CadD protein as efficiently as it does Cr^{3+} , Cu^{2+} , Zn^{2+} , and Cd^{2+} . This may be the reason for low removal activity of *E. coli/cadDET* for Pb^{2+} compared to other metal ions considered in the study.

cadD gene can be used to develop a biosorbent as CadD is a transmembrane protein, which has the ability of binding heavy metals (Crupper et al. 1999). Therefore, *cadD* gene along with a suitable promoter constitutively expressing CadD (to avoid the use of IPTG for induction), can be constructed as a biosorbent to have an enhanced heavy metal removal ability.

In conclusion, the isolated strain TWSL_22 was identified as *S. epidermidis* using 16S rDNA sequence and found to possess *cadA*, *cadD*, and *copA*. The developed recombinant *E. coli/cadDET* was able to express CadD (22 kDa) successfully and demonstrated significant metal removal ability and tolerance for various metal ions. The multimetal bioremoval and tolerance capacities of CadD reveal its potential to be used in the construction of biotechnological tools such as biosensors and biosorbents for *in vivo/ex vivo* detection and removal of Zn^{2+} , Cd^{2+} , Cu^{2+} , Cr^{3+} , and Pb^{2+} in industrial wastewater.

Ethics approval

Not applicable.

Consent for publication

The authors declare that they have no consent for publication.

Authors' contributions

C.D.W. and N.V.C. conceptualized, designed, and supervised the research. D.M.D.C.D. and W.M.N.H.K. conducted the experiments. D.M.D.C.D. analyzed data and wrote the manuscript. All authors read and approved the manuscript.

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Conflict of interest: The authors declare that they do not have any conflict of interests.

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